

VRAJKUMAR PATEL

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PROFESSIONAL SUMMARY

CS graduate (AI concentration) with production experience building full-stack and AI systems. Founded Stayvoo, a hotel booking platform, and built ARIA, an AI agent combining LLM inference and RAG. Led a 4-person engineering team at Squareone delivering full-stack features on AWS. 2nd place, NLU CSIS Tech Summit.

SKILLS

- **Programming:** Python, Java, JavaScript, HTML, CSS, MySQL, SQLite, React, Node.js, TypeScript, CI/CD, Restful API-, PostgreSQL, FastAPI, Django, C#/.NET, SQL, Pydantic, Bash/Shell scripting
- **AI/ML:** TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Large Language Models, Retrieval-Augmented Generation, Prompt Engineering, LLM API Integration, OpenCV / Tesseract, LlamaIndex, Vector databases, Deep Learning
- **Tools:** VS Code, PyCharm, Jupyter Notebook, Git, GitHub, Figma, Slack, Docker, Claude Code, AWS, Supabase, Railway, Playwright, n8n, Stripe API, Twilio API, Postman, Linux, Vercel, GitHub Actions
- **Security:** OWASP Security Principles, Authentication & Authorization, Rate Limiting, Audit Logging, IDOR Prevention, PII Protection, Penetration Testing

EDUCATION

National Louis University

Bachelor of Science, Computer Science & Information Systems | Artificial Intelligence.

Sep 2026

Chicago, IL

- **GPA:** 3.68

- **Achievements:** 2nd Place, CSIS Tech Summit - AI capstone project (ARIA), May 2026

WORK HISTORY

Stayvoo LLC

Jun 2026 - Present

Founder & Full-Stack Developer

Brookfield, WI

- Founded and built a hotel booking platform for extended-stay and group reservations, developing the full stack with FastAPI, React, and Supabase, deployed on Railway and Vercel
- Engineered an 8-phase security overhaul that fixed IDOR and PII enumeration vulnerabilities, added authentication, rate limiting, audit logging, and Playwright end-to-end test coverage, resulting in a more secure platform and preventing data-leak incidents
- Built an admin dashboard with React and Supabase that provides global search, inline editing, and a real-time daily operations view, allowing staff to manage reservations faster and reduce manual errors
- Designed automation architecture using n8n workflows, Twilio SMS, and Stripe payments with a feature-flag kill switch for controlled launch
- Configured a B2B lead-generation pipeline using Apollo.io targeting construction and staffing companies

Squareone

Jan 2026 - May 2026

Team Lead, Full Stack Software Developer Intern

Chicago, IL

- Led a 4-person team to build a full-stack job search application, planning sprints, assigning tasks, and mentoring teammates using Agile methods; the project was delivered on schedule and received positive feedback from the product owner
- Developed responsive frontend features with React and TypeScript and built scalable REST APIs with Django REST Framework and PostgreSQL, improving page load speed and enabling smooth data exchange across the platform
- Integrated AI-powered resume builder, review, and automated cover-letter generation, independently implementing the machine-learning models; these features increased user engagement by providing personalized application assistance
- Engineered backend systems including authentication, data modeling, and job management workflows, ensuring secure and efficient server-side functionality.
- Managed GitHub workflows, merged frontend and backend code, and supported AWS deployment, resulting in reliable end-to-end delivery of the application

PROJECTS

ARIA - Adaptive Responsive Intelligent Agent [Link](#)

- Built full-stack AI study companion with speech recognition, LLM inference, text-to-speech synthesis, and real-time animated avatar with emotional expressions.
- Engineered persistent memory system using Obsidian vault and BM25 search to enable AI to reference student's past projects and learning history
- Integrated NVIDIA Audio2Face for lip-sync animation and built emotion system where LLM outputs trigger 17 different avatar expressions synchronized to audio
- Optimized GPU performance across competing TTS, lip-sync, and animation models; achieved <5 second end-to-end latency on RTX 5000-series
- Tech stack: C# (.NET 9), Python, Groq LLM, ElevenLabs TTS, Qwen3 TTS, Live2D avatar, CUDA 12.8

On-Premise ID Scanning & Check-In Automation

IDscnr

- Developed a secure Windows application that scans IDs and passports using OCR, checks DNR lists, and auto-fills PMS systems, reducing guest check-in time to approximately 3 seconds while improving accuracy and front-desk security.

E-Commerce Customer Segmentation Dashboard [Demo](#)

- Built an end-to-end RFM and K-Means clustering pipeline in Python (Pandas, Scikit-learn) with interactive Matplotlib/Seaborn visualizations to segment e-commerce customers for targeted marketing.

Data-Driven Insights into Gun Violence for Community Intervention [Demo](#)

- Built a full Python analytics pipeline ingesting and correlating crime and socioeconomic data from multiple public sources, with automated validation checks and interactive visualizations surfacing intervention-relevant patterns.